

Layered Controls and Service Disruption in a Synthetic Hospital Network: A Paired Monte Carlo Analysis

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Abstract Public ransomware studies show what happened to hospitals, but they cannot show how the same disruption would have unfolded under a different set of controls. We built a Monte Carlo model of ransomware-like movement through a synthetic hospital network and measured the resulting loss of seven digital services. Baseline and layered-control runs shared the same topology, entry point, and event-level random values, so each scenario served as its own comparison. Our first 15-minute analysis favored the layered portfolio, but a diagnostic showed that the size of the effect changed with the simulation step. We therefore froze a new five-minute protocol and ran 500 fresh pairs. Mean weighted service-hours lost fell from 193.4 in the reference runs to 33.9 with basic segmentation, faster detection and isolation, and protected backups. The mean paired change was -159.5 hours (95% bootstrap interval, -166.1 to -152.7). Loss was lower in 451 pairs, unchanged in 48, and higher in one. The layered portfolio also had a lower mean in all 128 joint stress settings, although the median reduction was 28.7% and the fifth percentile was 3.1%. The model did not reproduce the multi-week disruptions reported in public studies. Our result is therefore about the direction of a synthetic comparison, not the real-world effectiveness of these controls. Hospital IT and clinical operations review would be needed before the model could support a local decision.

Keywords healthcare cybersecurity; ransomware; Monte Carlo simulation; common random numbers; service continuity; model verification; sensitivity analysis

1 Introduction

Counting infected computers misses the part of a hospital cyberattack that matters most to patients and staff: which services still work, and for how long. In 374 publicly reported attacks on U.S. healthcare delivery organizations from 2016 through 2021, 44.4% mentioned disruption to care and 41.7% mentioned electronic-system downtime. For incidents with a reported duration, the mean was 15.8 days [1]. Those numbers describe a selected set of visible attacks; they are not a hospital's probability of being attacked.

The operational record is becoming clearer. WannaCry was associated with lost appointments and reduced activity at affected National Health Service trusts [2]. During a month-long U.S. attack, two neighboring emergency departments handled more patients and ambulances and experienced longer waits and stroke-care delays across 19,857 visits [3]. Later work reported changes in cardiac-arrest incidence and outcomes at nearby, unaffected hospitals [4]. Using a national attack database linked to Medicare claims, Neprash et al. [5] estimated a 17–24% decline in hospital volume during the first attack week, with recovery over roughly three weeks. The studies differ in design and outcome, but they agree on one practical point: digital outages can become clinical and regional operating problems.

CISA, HHS, and NIST recommend preparation, segmentation, detection and response, identity protection, and recoverability [6–8]. Their guidance does not tell a modeler what number to use for a spread probability or isolation-success rate, nor should it. We treated those quantities as assumptions to stress, not as measurements borrowed from guidance.

Our question was: *when the same synthetic incident is replayed under a flat reference posture and a feasible layered portfolio, does the layered portfolio preserve more modeled service availability across new scenarios, joint parameter changes, and different time steps?* We expected the answer to be yes. The useful test was whether that answer survived less favorable assumptions and whether its size remained stable. It did survive; its size did not. The model is not an estimate of hospital risk, patient harm, or the causal effect of deployed controls.

1.1 Related empirical and simulation research

Downtime research helps connect unavailable software to work on the hospital floor. Among 80,381 patient-safety reports, Larsen et al. [9] found 76 that described EHR downtime; laboratory and medication processes appeared often, and downtime procedures were not used consistently. A mixed-methods study of a roughly 48-hour outage found delays in staff accounts that were not fully visible in electronic timestamps [10]. Related AHRQ work used records, observation, interviews, and simulation to study laboratory and emergency-department contingencies [11]. This evidence is why our outcome is service availability rather than only the number of compromised nodes. It also makes clear what our binary service measure leaves out.

The closest simulation studies operate at different layers. A patient-flow model compared ransom payment and backup restoration across an emergency department, intensive care unit, and laboratory [12]. A hybrid process model examined how device outages and emergency plans affect trauma care [13]. Other work has modeled financial loss with bond percolation [14] and infrastructure interdependencies with agents [15]. Most recently,

Arsalani et al. [16] combined data from two Alabama incidents with a regional EMS simulation and tested the role of hospital IT segmentation. That paper is especially relevant: it means we cannot claim to be the first to simulate ransomware, hospital resilience, or segmentation.

Table 1 states the remaining distinction without a priority claim. Our model stays inside one synthetic hospital and links directed digital propagation to seven service dependencies. It compares portfolios with stable, event-indexed random values, varies 13 assumptions together, checks the result at several time steps, and retains a failed diagnostic. No single item is new on its own. The contribution is the combination and its auditable execution.

Table 2 shows how public evidence was used without converting observed operational outcomes into unsupported cyber-transition rates.

2 Methods

2.1 Model scope

The Python model is defensive: it contains no malware, exploit code, scanning, or connection to a real network. A generated directed graph represents permitted communication or access paths among workstations, EHR, laboratory, pharmacy, imaging, medical devices, administration, identity, backup, and internet-facing/vendor functions. Nodes move among healthy, compromised, detected, isolated, restoring, and restored states. Propagation can occur only along a directed edge and is modified by vulnerability, patch status, privilege, segmentation, identity compromise, backup access, and the selected portfolio.

Three neutral network scales (40–60, 200–300, and 750–1,000 nodes) and three synthetic postures were used in a cross-setting replication. These labels do not correspond to real hospital size, geography, or maturity. Entry types were a workstation, internet-facing service, privileged system, medical device, or vendor connection.

2.2 Service outcomes

The primary outcome was weighted service-hours lost across EHR, laboratory, pharmacy, imaging, scheduling, identity, and backup/recovery. At each step, a service was unavailable if its core was nonfunctional, fewer than 60% of its support nodes remained functional, or a required identity dependency was unavailable. Functional means neither actively compromised nor isolated, so defensive isolation can itself remove service capacity.

Reference weights were 1.0, 0.8, 0.9, 0.7, 0.4, 0.9, and 0.6, respectively. The 60% rule and weights are value judgments, not clinical standards. We therefore retained unweighted, service-specific downtime in every row and varied the functional threshold from 45% to 75% in joint stress tests. A secondary sustained-outage indicator was set when any of four clinical services was unavailable for more than two modeled hours. We avoid the earlier term “catastrophic risk” because the indicator is neither clinically validated nor a real-world probability.

2.3 Evidence and parameter provenance

Every influential input was classified as empirical observation, evidence-informed window, outcome definition, or stress assumption. Public studies informed the 72-hour acute window and the need for 7- and 21-day recovery checks. Mechanistic values such as per-edge spread, patch multiplier, detection delay, isolation success, identity escalation, restoration throughput, false-positive isolation, legacy fraction, and residual backup access were not identifiable from those studies. They remained synthetic stress variables.

The final joint design varied 13 assumptions with Latin-hypercube sampling: 15-minute-equivalent spread 0.05–0.50; patch effectiveness 0.30–0.95; within-zone degree 2–5; cross-zone degree 1–4; identity multiplier 1–2; restoration rate 0.25%–4% of nodes per 15 minutes; no-backup restoration multiplier 0.10–0.50; service threshold 0.45–0.75; false-positive isolation 0–0.002 per node per 15 minutes; legacy fraction 0.05–0.35; isolation success 0.25–0.95; detection delay 0.75–24 hours; and residual isolated-backup traversal 0–0.10. Clock-time hazards and durations were converted for the five-minute replication. None of these ranges is an estimate of a sector-wide hospital parameter.

2.4 Portfolios and pairing

The reference used a flat graph, the posture’s baseline patch and response values, and connected backups. The layered portfolio combined basic segmentation, a one-tier detection improvement, rapid same-step isolation attempts, and protected backups. “Protected backup” is an abstraction that compresses credential separation, management-plane access, immutability, testing, reconnection, and reinfection into a small set of paths; residual access was explicitly stressed.

All comparisons used common random numbers, but not by simply restarting a stateful random-number generator with the same seed. That shortcut can lose event alignment when an intervention changes the code path [18]. Instead, a stable key made from the scenario, event type, and time step generated a full node- or edge-level random vector. A portfolio could change which entries were used, but it could not shift the values assigned to events shared with the other portfolio. The generated base topology, entry type and node, and patch draws were also held fixed within a pair. Thus a portfolio did not receive an easier network simply by chance.

2.5 Study sequence and sample size

Earlier repository results were treated as development evidence and were not pooled with the new study. Before generating the 15-minute validation, we froze a local protocol, configuration, seeds, and file hashes. Planning data gave a paired-difference standard deviation of 73.90 service-hours and a reference mean of 169.97. Requiring Monte Carlo standard error no greater than 2% of that reference mean gave 473 pairs; we rounded to 500.

The 15-minute phase contained 500 primary pairs, 100

Table 1: Closest simulation research and the scope of the present study. The comparison identifies differences in modeled layer and outcome; it is not a claim that earlier studies are incomplete.

Study	Modeled layer	Main comparison or outcome	Difference from this study
Ghayoomi et al. [12]	Patient flow and constrained hospital resources	Ransom payment versus restoration strategies	We model digital asset propagation and service availability, not patient throughput.
Willing et al. [13]	Trauma-room process and device failures	Security measures and emergency plans	We compare network-control portfolios across synthetic hospital services.
Chiaradonna et al. [14]	Mixed random graphs	Aggregate cyber-loss distributions	Our endpoint is time-varying digital-service loss, not financial loss.
Carramiñana et al. [15]	Interdependent infrastructure agents	Node availability and quality of service	We use paired within-scenario portfolio differences and explicit numerical diagnostics.
Arsalani et al. [16]	Regional hospitals and ambulance operations	EMS impact and hospital IT segmentation	We model within-hospital propagation; regional demand and ambulance operations are absent.
Present study	Directed synthetic assets and seven dependent services	Flat reference versus layered controls under joint uncertainty	Event-indexed pairing, metric reconciliation, and time-step failure are reported together.

Table 2: Evidence used to define the study boundary. Operational studies support outcomes, time windows, and validation patterns; they do not identify the model’s universal transition probabilities.

Evidence	Observation used	Permitted modeling use
Reported U.S. incidents [1]	Disruption was common among reported attacks and sometimes lasted weeks.	Test multi-day horizons and avoid treating a 48- or 72-hour run as complete recovery evidence.
WannaCry impact [2]	Admissions, appointments, and activity changed at affected trusts.	Prefer service and operations outcomes over machine counts.
Adjacent-hospital studies [3, 17]	Disruption can shift demand to neighboring facilities.	Identify regional spillover as missing model structure, not as a fitted parameter.
Downtime case research [9–11]	Manual workarounds, reporting gaps, and process delays matter.	Report binary service availability as a simplification and preserve service-specific outcomes.
Hospital simulation studies [12, 13]	Patient flow and resource constraints can connect digital outages to care processes.	Define a future patient-flow layer; do not claim the present asset model measures patient harm.
Government guidance [6–8]	Layered controls and recoverability are recommended.	Define relevant portfolios and failure modes, not numerical efficacy.

scenarios in each of nine scale-by-posture combinations, and 128 joint stress settings with 20 scenarios each. Pre-specified 5-, 15-, and 30-minute diagnostics initially revealed a scenario-pairing and clock-time conversion error. Original files were retained, the deviation was documented, and only those diagnostics were corrected. The corrected diagnostic then showed material time-step sensitivity. An explicitly exploratory 1-, 2-, and 5-minute pilot was run. Finally, a post-diagnostic five-minute protocol was frozen before 500 new primary pairs and 128 new stress settings with 10 scenarios each. It used fresh seeds and scenario identifiers.

Table 3 accounts for every new execution, including superseded diagnostics. A run is one portfolio under one synthetic scenario, not a hospital or incident.

Table 3: Fresh execution accounting. Superseded diagnostic rows remain in the archive and are never pooled with corrected results.

Phase	Runs
15-minute primary, cross-setting, joint stress	11,880
Initial diagnostics (retained)	1,800
Corrected diagnostics	1,800
Exploratory fine-step pilot	300
Frozen five-minute replication	3,560
Total	19,340

2.6 Analysis and verification

The primary statistic was the within-scenario difference (layered minus reference) in weighted service-hours. We report means, medians, the distribution of paired changes, and 95% percentile bootstrap intervals that re-sample complete scenarios. Joint-stress results were first averaged within parameter setting; robustness required a negative primary interval, lower mean loss in at least 95% of settings, and a positive fifth percentile of setting-specific relative reduction. This success rule concerns model behavior only.

Fifty-six unit and invariant tests and twelve behavioral validation cases covered transitions, pairing, seed reproduction, segmentation, patching, isolation, backup behavior, dependencies, budgets, and export. An independent script recomputed all three derived ratios from raw numerators, denominators, and final-service flags. The study follows the transparency intent of the STRESS reporting guideline [19]; protocols, parameter registers, deviations, raw outputs, and analysis manifests are retained.

3 Results

3.1 Five-minute paired replication

Across 500 fresh pairs, reference scenarios averaged 193.4 weighted service-hours lost and layered scenarios averaged 33.9 (Table 4). The paired mean change was -159.5 hours (95% bootstrap interval, -166.1 to -152.7), and its Monte Carlo standard error was 3.42, below the protocol target of 3.87. The layered portfolio produced lower loss in 451 scenarios, tied in 48, and produced 11.35 more hours in one scenario. The mean-of-groups reduction was 82.5% *within the synthetic model*; this percentage is not a real control-effectiveness estimate.

The reference sustained-outage indicator was positive in 90.2% of scenarios and the layered indicator in 41.2%, a paired difference of -49.0 percentage points. These are conditional model frequencies under a deliberately severe scenario bank, not incident probabilities.

Service-specific results did not depend on weights for their direction. Mean EHR downtime changed from 32.9 to 7.4 hours, laboratory from 36.8 to 7.3, pharmacy from 34.7 to 6.9, imaging from 40.7 to 7.6, scheduling from 49.3 to 8.0, identity from 32.1 to 6.2, and backup/recovery from 38.0 to 0.6 *within the synthetic model*. These values describe binary modeled availability, not hours of complete clinical shutdown.

3.2 Joint uncertainty

The layered portfolio had lower mean loss in all 128 five-minute joint stress settings (Wilson 95% interval for the favorable fraction, 97.1%–100%). The median setting-specific reduction was 28.7%, the fifth percentile was 3.1%, and the 95th percentile was 96.4% (Figure 2). The success rule was met, but the low fifth percentile means the advantage was small in some plausible stress combinations. In the earlier 15-minute stress bank, the median

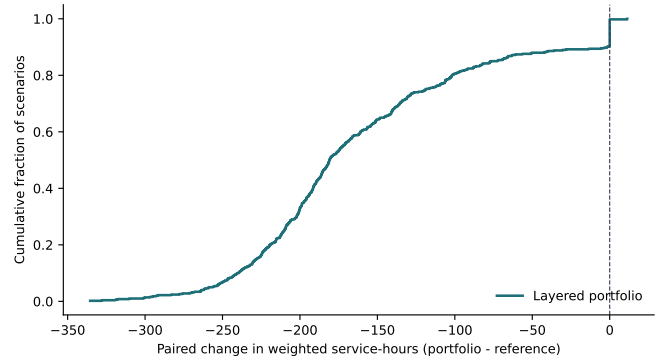


Figure 1: Distribution of paired five-minute effects. Negative values favor the layered portfolio. Most scenarios improved substantially, 48 tied at zero, and one was worse; reporting the complete distribution prevents the mean from hiding that exception.

and fifth percentile were 66.5% and 39.0%, respectively. The direction replicated; the magnitude did not.

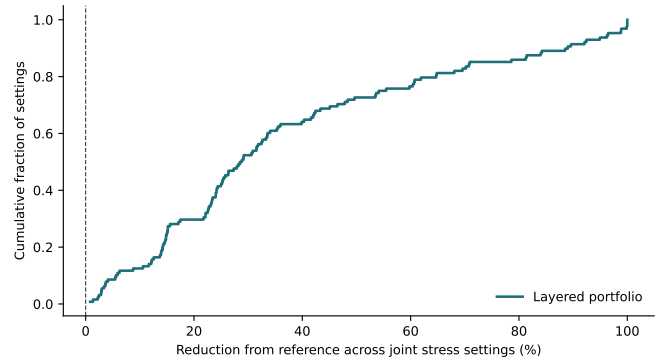


Figure 2: Five-minute joint stress results. Every setting favored the layered portfolio, but reductions ranged from nearly zero to very large. The lower tail, not the best case, limits the strength of the conclusion.

3.3 Cross-setting and numerical checks

In the independent 15-minute cross-setting bank, the layered mean was below the reference mean in all nine scale-by-posture cells. Reductions of the cell means ranged from 55.2% to 94.2%. Scenario-level improvement fractions were lower in stronger-posture cells because many reference scenarios had little or no loss, creating ties. These labels remain synthetic; the result does not show that a particular real hospital category would benefit by those percentages.

The corrected coarse-step diagnostic produced layered/reference reductions of 81.7%, 88.4%, and 94.0% at 5, 15, and 30 minutes. The exploratory fine-step pilot produced 82.3%, 82.6%, and 85.6% at 1, 2, and 5 minutes. Paired mean differences in that pilot ranged from -155.8 to -175.0 hours with Monte Carlo standard errors near 10 hours. Thus, fine resolutions gave a broadly similar comparative answer, but neither absolute values nor exact percentages were numerically invariant. The 30-minute step was especially optimistic for the layered portfolio because fewer propagation generations can occur before a fixed clock-time response.

Table 4: Primary five-minute replication. The paired interval and scenario counts support the direction of the comparison; absolute hours and the 82.5% ratio remain conditional on the model and time step.

Comparison		Pairs	Reference mean	Layered mean	Paired change (95% CI)	Lower / tie / higher
Weighted hours	service-	500	193.4	33.9	-159.5 (-166.1, -152.7)	451 / 48 / 1
Sustained-outage indicator	indicator	500	90.2%	41.2%	-49.0 percentage points	-

At the 15-minute reference settings, 95% of flat-reference scenarios recovered within 72 hours and 100% within seven days. This is shorter than the multi-week disruptions documented in public incident reports [1]. The model therefore fails that empirical duration pattern under its reference recovery values. We do not interpret modeled recovery time as a prediction.

3.4 Derived-metric reconciliation

Anne Lee’s review noted that three secondary ratios had been excluded without a satisfactory explanation. Independent reconstruction showed that the archived `pct_compromised` field equals ever-compromised nodes divided by total nodes, and `pct_clinical_capacity_lost` equals time-averaged weighted unavailability of four clinical services. Both agreed across all 15 archived CSV files to within 5.0×10^{-11} .

The third name, `pct_services_restored`, was misleading: it measured the fraction of all seven services available at the final step, including services never disrupted. Legacy rows did not store the seven final flags, so the field was not independently row-auditable; exact reruns of sampled archived rows nevertheless matched. New rows retain the historical alias for compatibility but add explicit final flags, `final_service_availability_fraction`, and `disrupted_service_recovery_fraction`. Independent recomputation of all new validation files agreed to within 5.0×10^{-11} . No headline analysis uses the ambiguous legacy names.

4 Discussion

4.1 What the comparison showed

Our first result looked almost too clean. At a 15-minute step, the layered portfolio’s median reduction across stress settings was 66.5%. Repeating the study at five minutes left the direction unchanged—all 128 settings still favored the layered portfolio—but reduced the median to 28.7%, with the lower tail close to zero. That diagnostic changed how we interpret the paper. The 82.5% reduction in the primary scenario bank is a description of those runs, not a stable estimate of control effectiveness.

The direction is less surprising than the amount. We encoded segmentation to remove paths, faster response to shorten the time available for spread, and protected backups to preserve a recovery route. The useful observation is that their combined advantage remained after

13 assumptions were varied together and after the simulation step was shortened. This agrees with layered-defense guidance [6–8]; it does not show that our numerical multipliers match controls deployed in a hospital.

4.2 How the workflow could be used

We see a possible use in tabletop planning, but only after external review and local calibration. A hospital could describe service dependencies without recording sensitive architecture, replace our synthetic ranges with authorized summaries, and define degraded operation with clinical staff. The output should be a short robustness report: which portfolios keep their ranking, which assumptions reverse it, and which missing measurements matter most. It should not be treated as a substitute for a risk assessment, architecture review, incident-response plan, regulatory obligation, or procurement process.

4.3 Where the model breaks down

We do not have a dataset that measures hospital topology, permitted paths, attack transitions, detection and containment, restoration capacity, service dependencies, and control costs in the same incidents. As a result, the confidence intervals cover Monte Carlo variation inside our model and nothing more. They do not quantify structural error or transferability to a hospital.

The time-step diagnostic is the clearest technical weakness. Finer steps allow more rounds of propagation before the same clock-time response, and they change the order of events. Five-minute runs reduced the discrepancy but did not make the absolute results invariant. We would replace this mechanism with an event-driven continuous-time model before interpreting exact hours or percentages.

Service availability is binary and depends on one support threshold. Real departments operate in degraded modes using paper, phones, manual verification, reduced menus, and workarounds whose effectiveness changes with time and staff load [9, 10, 20]. Because our model does not contain patient flow, the phrase “service-hours lost” cannot be translated into patients delayed, diverted, or harmed. Regional spillover observed by Dameff et al. [3] and Abouk and Powell [17] is also absent.

Synthetic postures bundle patching, detection, isolation, legacy assets, segmentation, and backups. They are scenario devices, not validated maturity classes. “Protected backups” simplify credentials, management planes, immutability, testing, reconnection, and reinfection.

tion. Normalized budget points are not dollars. The portfolio search cannot discover a control or failure mode that was not encoded.

Public reports create another problem: visible incidents are selected, and “recovery” may refer to technical restoration, clinical workflow, billing, or normal operations. Even with that caution, our reference model recovered far sooner than the multi-week pattern reported in the literature. We treat that mismatch as evidence that its recovery output should not be used predictively.

4.4 Next validation step

This paper stops before outside contact. The next step is a structured review by hospital IT or incident-response staff and by clinical-operations staff, with privacy and research-ethics preparation completed first. We would ask about state transitions, degraded service, dependencies, timing, backup failure modes, and useful outputs. We would not request network diagrams, credentials, vulnerability details, patient data, or active-incident information. A review of this kind would establish face validity at most. Predictive validation would still require suitable withheld incident or exercise data.

5 Conclusion

In our fresh paired runs, basic segmentation, faster detection and isolation, and protected backups reduced modeled service disruption relative to a flat reference. The ranking survived every joint stress setting at both 15- and 5-minute resolution. The estimated size did not. At five minutes, some stress settings showed only a small advantage, and absolute outcomes still changed with numerical resolution.

We are therefore confident about one result inside this model: the layered portfolio was directionally robust over the uncertainty we tested. We are not confident that the hours or percentages transfer to a hospital. The metric repair, retained diagnostic error, and failed recovery-duration check are part of the result because they show where the analysis stops being reliable. Operational review and a continuous-time model with degraded-service states are needed before using the framework for a real decision.

Research integrity and declarations

Ethics. The study used synthetic networks, simulated trajectories, and public literature. It involved no participants, patient records, private organizational data, malware, or intervention. External interviews have not been conducted. Appropriate school, competition, institutional, consent, and privacy requirements should be checked before collecting reviewer data.

Data and code. The earlier development repository is at <https://github.com/Binded-Person/NazeResearchNSRI64>. The revised code, protocols, configurations, manifests, raw outputs, processed summaries, tests, and figures are included in the accompany-

ing local research package. At the authors’ request, this revision has not been pushed or published. A versioned public release should accompany any submission.

Artificial-intelligence use. We used Claude, ChatGPT, and Codex substantially while developing and revising this project. Their roles included code suggestions, debugging, search assistance, analysis checks, organization, and sentence-level editing. We did not use their output as a source. Numerical claims were regenerated from code and CSV files, and literature claims were checked against the cited papers or official documents. The tools also helped us find the time-step diagnostic error, which we retained in the study record. Before submission, each author must be able to explain the model, reproduce the main analysis, and approve the wording. The authors remain responsible for the study and for any errors that remain.

Author contributions. A final CRediT contribution statement has not been assigned in this draft and must be completed by the authors without inflating any person’s role.

Funding. The authors report no specific funding for this work.

Conflicts of interest. The authors report no known competing interests; all authors should reconfirm this statement before submission.

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